

Genomic selection in plant breeding: Key factors shaping two decades of progress

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ABSTRACT

Genomic selection, the application of genomic prediction (GP) models to select candidate individuals, has significantly advanced in the past two decades, effectively accelerating genetic gains in plant breeding. This article provides a holistic overview of key factors that have influenced GP in plant breeding during this period. We delved into the pivotal roles of training population size and genetic diversity, and their relationship with the breeding population, in determining GP accuracy. Special emphasis was placed on optimizing training population size. We explored its benefits and the associated diminishing returns beyond an optimum size. This was done while considering the balance between resource allocation and maximizing prediction accuracy through current optimization algorithms. The density and distribution of single-nucleotide polymorphisms, level of linkage disequilibrium, genetic complexity, trait heritability, statistical machine-learning methods, and non-additive effects are the other vital factors. Using wheat, maize, and potato as examples, we summarize the effect of these factors on the accuracy of GP for various traits. The search for high accuracy in GP—theoretically reaching one when using the Pearson's correlation as a metric—is an active research area as yet far from optimal for various traits. We hypothesize that with ultra-high sizes of genotypic and phenotypic datasets, effective training population optimization methods and support from other omics approaches (transcriptomics, metabolomics and proteomics) coupled with deep-learning algorithms could overcome the boundaries of current limitations to achieve the highest possible prediction accuracy, making genomic selection an effective tool in plant breeding.

Key words: genomic selection, genetic gain, genomic prediction optimization, deep learning, training population optimization

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INTRODUCTION

Global population growth is likely to continue at a similar or faster pace in the coming decades. Demand for food is expected to increase by the same amount to feed the population while crop productivity has been curtailed by various biotic and abiotic stresses exacerbated by anthropogenic climate change. Plant breeding is fundamental to developing new cultivars with higher yield,

improved quality, and tolerance or resistance to several abiotic and biotic stresses. For example, wheat production at the global level has increased from 200 million tons in 1961 to 775 million tons in 2023 (FAO, 2023) with no significant change in total area

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of wheat production (220 million hectares). This is principally due to the development and deployment of semi-dwarf high-yielding and input-responsive new wheat cultivars (Borlaug, 2002) with resistance and tolerance to major biotic and abiotic stresses, respectively, along with improved agronomic management, mechanization, favorable policies, and infrastructures across the entire wheat value chain (Tadesse et al., 2019).

Genetic enhancement of crops has long relied on conventional cross-breeding methods whereby breeding and selection of genotypes are solely based on pedigree and phenotypic performance. Rigorous evaluation of parents for different traits, targeted crossing, generation advancement using the summer and winter shuttle breeding schemes to shorten the breeding cycle, key location evaluation of elite germplasms, and effective database management have played significant roles in developing improved crop cultivars. However, the expeditious emergence of DNA-sequencing technology has allowed breeders to gain comprehensive genomic information on crops, which is very valuable for selection. The development of several DNA-marker-based genotyping systems significantly increased the number of DNA markers available to plant breeders (Crossa et al., 2017). This breakthrough allowed plant breeders to select plant performance based on their genetic marker composition rather than solely on their phenotypic performance, which is prone to several limitations in selection efficiency.

The application of genomic tools in the breeding practice of plants, generally termed genomic-assisted breeding, has progressed through various stages in the last four decades (Varshney et al., 2021). It started with linkage-based mapping of quantitative trait loci (QTLs) (Soller and Plotkin-Hazan, 1977) where, with a limited number of DNA markers, those segregating with a particular trait were identified as linked to a QTL and used for marker-assisted selection (MAS). The method required a set of segregating individuals developed from biparental crosses, a time-consuming procedure, with a narrow allelic variation and poor resolution that leads to low impact in practical plant-breeding programs (Bernardo, 2008). The genome-wide association study (GWAS) approach became a popular and powerful method for identifying markers closely linked to QTLs of target traits (Zhu et al., 2008; Tibbs Cortes et al., 2021). However, the practical implementation of the method via MAS has been constrained to limited numbers of major QTLs while numerous small-effect QTLs in complex traits have remained unknown and unutilized (Jannink et al., 2010).

Genomic selection (GS), when developed GP models are applied in practical selection, has emerged as a powerful tool in plant breeding, particularly after the advancement of readily available genome-wide single-nucleotide polymorphisms (SNPs). Besides early contributors (Lande and Thompson, 1990; Bernardo, 1994; Nejati-Javaremi et al., 1997; Haley and Visscher, 1998; Whittaker et al., 2000), GS was first elaborated two decades ago by Meuwissen et al. (2001). In this groundbreaking study, the authors paved the way to a new avenue in plant breeding, suggesting that prediction of genetic values from marker profiles could extensively increase genetic gain in plant and animal breeding, particularly if combined with reproductive techniques to shorten the generation interval. The conventional MAS approaches tend to focus solely on a limited set of markers linked

with well-investigated major QTLs excluding the vast majority of minor-effect QTLs. In contrast to these methods, GP employs large number of genome-wide SNPs to quantify the comprehensive genetic merit of individual plants encompassing most contributing QTLs of a target trait (Bernardo and Yu, 2007; Heffner et al., 2009). The continued rapid advancement of next-generation sequencing technology to produce dense genome-wide SNP markers, coupled with its substantial cost reduction for genotyping in several crops, makes GS a must-implement method in most breeding programs. Empirical research has shown the advantage of GS for accelerating the genetic gains per unit of time over pedigree-based selection. GS has emerged with huge potential to reduce the cost per breeding cycle, increase selection intensity and accuracy, and significantly reduce the time required to develop a cultivar compared to phenotypic-based selection (Crossa et al., 2010, 2017; Edwards et al., 2019).

Developing statistical machine-learning models and training population optimization are the two main thematic areas actively explored in plant GP research. This is because of their potential to improve the prediction accuracy while the current achievement is far from optimal. This review begins with a simplified explanation of GP followed by an exploration of the up-to-date widely applied cross-validation (CV) methods in plant breeding. After a comprehensive overview, details of the key factors affecting GP accuracy identified over the last two decades are elaborated. Moreover, empirical research results are analyzed using wheat, maize, and potato as examples of self-pollinating, cross-pollinating, and clonally propagated crops, respectively, to illustrate the impact of the identified factors on the accuracy of GP in various traits. Finally yet importantly, the implementation of GS is highlighted in a showcase example from ongoing empirical studies from public and private breeding programs. In summary, valuable suggestions are forwarded to support the successful implementation of GS in plant-breeding programs.

GENOMIC PREDICTION

GP is the most recent data-driven method that has been widely accepted and used as a valuable tool to accelerate genetic gain in plant-breeding programs (Desta and Ortiz, 2014; Bassi et al., 2016; Xu et al., 2020). GP employs advanced statistical machine-learning models to select individuals within a breeding population based on breeding values estimated from genome-wide markers. This selection process relies on data from a training population, encompassing both phenotypic and genotypic information (Figure 1A). After a rigorous training procedure, these models generate predictions of breeding or phenotypic values for traits of a target population consisting only of genotypic data. However, the performance of prediction models should be first evaluated through CV before applying selection (see the next section for details of CV methods). This step in GP is critical in order to evaluate the performance of prediction models and compare different sets of statistical machine-learning models with various scenarios, such as incorporating multiple traits, known major genes and marker-trait associations (QTLs), genotype $\times$ environment (G $\times$ E) interaction, and other omics data such as transcriptomics, metabolomics, and proteomics (Figure 1A).

Comparisons among GP methods are evaluated through their prediction accuracy, which is directly linked to the breeder's

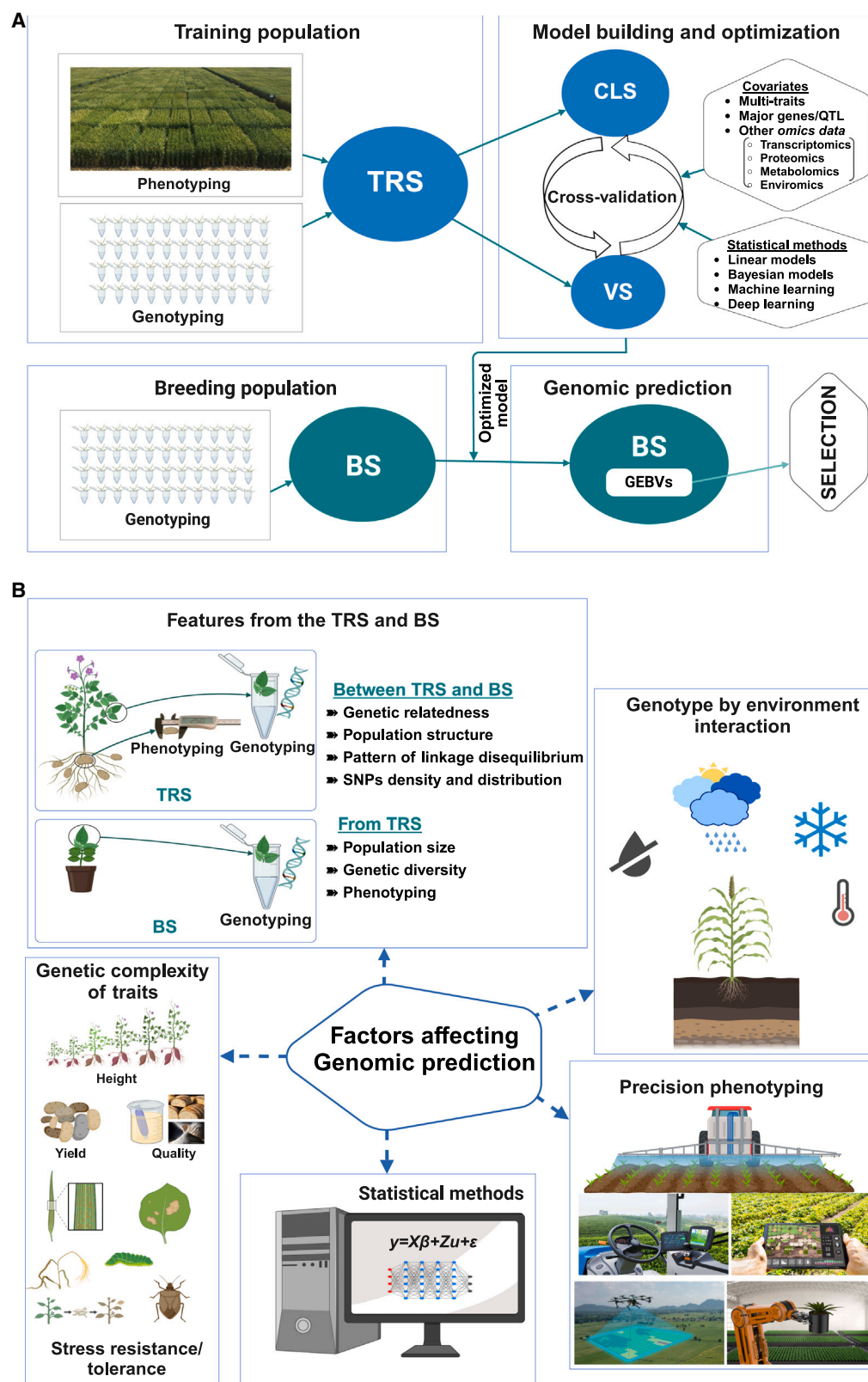


Figure 1. Schematic overview of GP model building and optimization, and major factors affecting genomic prediction.

(A) In genomic prediction, phenotypic and genotypic data as well as other covariates can be applied to develop and optimize various machine-learning methods splitting the optimized training population into calibration and validation sets and estimating the prediction accuracy through cross-validation.

(legend continued on next page)

equation (Akdemir and Isidro-Sánchez, 2019). Various factors can affect GP, and the accuracy score varies significantly across experiments for a single trait. For instance, the prediction accuracy of a single trait in wheat, maize, and potato hugely varied across different experimental research due to the different setups in training population composition, applied statistical machine-learning models, and other factors (Supplemental Tables 1–3). The GP accuracy (r_{MG} [correlation between marker predicted value with true predicted genetic value]) is measured as the Pearson's correlation between genomic estimated breeding value (GEBV) and true breeding value (Combs and Bernardo, 2013; Isidro et al., 2015), which gives an estimate of selection accuracy (Merrick et al., 2022). Selection accuracy is directly related to selection response (R), also known as genetic gain, and in the breeders' equation is calculated as $R = i r \sigma_A / t$, where i and r are the selection intensity and accuracy, respectively; while σ_A is the square root of the additive genetic variance and t is the cycle time (Falconer and Mackay, 1996).

GP considers the breeding values of parental average and deviation of Mendelian sampling to define GEBVs of an offspring, which allows the method to be used for: (1) rapid selection cycle with short breeding interval at early generations via prediction of the additive effects (i.e., GS at the F_2 level of a biparental cross); and (2) selection of lines at late stages of selection by predicting the genotypic values of individuals, with both additive and non-additive effects determining the final commercial value of the lines (Crossa et al., 2014; Dreisigacker et al., 2023).

Numerous factors affect GP and can significantly reduce its accuracy (Figure 1B). Consequently, unless adequately addressed, they can hinder the effective utilization of GP in plant-breeding programs. The population size, genetic diversity, and genetic relatedness with the breeding population are key features to target during training population optimization. Factors such as the level of linkage disequilibrium between QTLs and markers (in both the training and breeding [testing] population), genetic complexity and heritability of target traits, quality/precision phenotyping, statistical machine-learning models, $G \times E$ interaction, and other non-additive factors are the other major features that further complicate GP in plant breeding.

CROSS-VALIDATION METHODS

CV is a fundamental technique in statistical machine-learning methods that aids model evaluation, hyperparameter tuning, and ensuring robust model performance. It plays a crucial role in building models that can make accurate predictions on new,

unseen data while avoiding overfitting and data-specific biases. GP models should initially be evaluated using CV methods before applying for the selection of candidate individuals in the breeding population. CV simulates the model's prediction performance by dividing the training population (training set; TRS) into calibration and validation sets.

Different GP CV methods are utilized depending on various determining scenarios (Figure 2). The K-fold CV is one of the most widely applied methods, where the entire dataset is divided into an equal number of folds. In the 5-fold CV method, for example, the TRS dataset is randomly grouped into 5-folds and prediction models are trained using the 4-folds as a calibration set while the remaining fold is used as a validation set. The accuracy could be measured after either averaging multiple runs from each fold or averaging runs comprising all folds. Leave-one-out CV (LOOCV) is the other method in which a single genotype is excluded from the calibration set and used as the validation set in each single iteration. An equal number of CV iterations are required with the number of samples or genotypes in this method. Hence, LOOCV is computationally intensive and only suitable for few genotypes (samples), while the 5-fold CV method is ideal for large datasets (Cheng et al., 2017). The other CV scenario has arisen in the case of multi-environment GP analysis (Crossa et al., 2017). Cross-validation 1 (CV1) is a scenario in which the GEBVs of newly developed lines or varieties are predicted in tested environments, thus being CV1 appropriate for predicting untested lines in tested environments. CV2, also known as sparse testing, is a method for genotypes tested in some environments and predicted in other tested environments. For this reason, CV2 is a reasonable option for predicting tested lines in tested environments. The other scenarios are CV0, which arises from the prediction of tested genotypes in an untested (unobserved) environment, while CV00 is used for predicting GEBVs of untested genotypes in unobserved environment (Figure 2).

TRAINING POPULATION

A TRS is used to establish the statistical relationship between genetic markers and phenotypic data for target traits to predict the phenotypic performance of individuals from their genotypic profile. In GP, the TRS should first be optimized to enhance the prediction accuracy and efficiency in breeding programs (see "training population optimization"). The optimized TRS can be of two types during the GP model optimization and application in the practical selection scenario. The first type is the parcel of the optimized TRS (calibration set) used to train the prediction models and estimate the GEBVs of the remaining individuals within the TRS (validation set) via CVs (Figure 1A). The second type is the overall optimized TRS applied to train the optimized GP models in a

The optimized model with the highest possible prediction accuracy is identified and applied to predict GEBV of the breeding population followed by selection of individuals based on their genetic merit for target traits.

(B) Various factors affect genomic prediction accuracy in plant-breeding programs. These factors arise from diverse sources at different stages during the analysis. Population size and genetic diversity of the training population, genetic relationship (kinship) and population structure of the training population with the breeding population, and quality of the phenotypic data applied in the statistical machine-learning models are features connected with training population and should be optimized during TRS development. Other factors including density and distribution of genetic markers across chromosomes, level of linkage disequilibrium between QTL alleles and marker alleles, genetic complexity and heritability of target traits, applied statistical methods, and non-additive genetic factors such as genotype-by-environment ($G \times E$) interactions hugely affect the final output of the GP accuracy. TRS, training population; BS, breeding population/set; CLS, calibration set; VS, validation set; GEBV, genomic estimated breeding value. All figures are created with BioRender (<https://biorender.com/>).

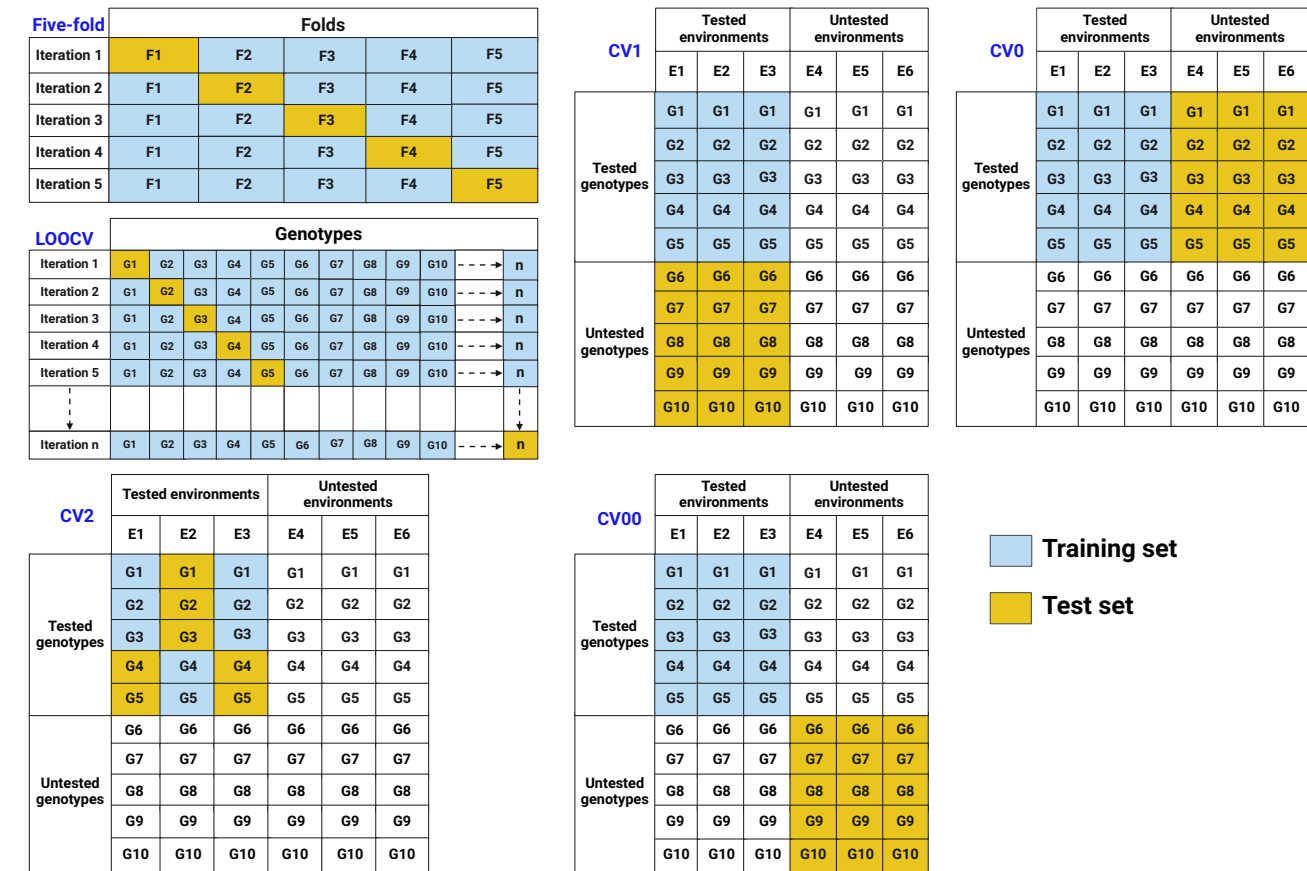


Figure 2. Genomic prediction cross-validation methods in plant breeding.

With the 5-fold cross-validation method, the complete population is initially allocated at random to 5-folds (F5). The 4-folds are then used as a calibration set in order to develop the GP model while retaining the remaining one as a validation set. A single genotype is excluded from the calibration set in the LOOCV, and its GEBV is predicted in every iteration. In multi-environment GP, a newly developed untested genotype can be predicted in tested environments (CV1), a genotype tested in some environments but untested in others (also known as sparse testing [CV2]), tested genotype predicted in an untested environment (CV0), and an untested genotype in an untested environment (CV00).

practical breeding scenario to estimate the GEBVs of individuals in the breeding population/set (BS), which are ready for selection. Features of the TRS including the population size, genetic diversity and genetic relatedness with the BS, population structure, level of linkage disequilibrium (LD) related to the BS, and the quality of phenotypic and genotypic data significantly affect the GP accuracy (Pszczola et al., 2012; Crossa et al., 2014; Hickey et al., 2014; Zhang et al., 2017a; Edwards et al., 2019).

Sample size of the training population

The ultimate goal of plant breeders is to achieve highly accurate but inexpensive estimates of genetic value (Lorenz and Nice, 2017). In GP, increasing the TRS size could inflict both positive and negative consequences for successful implementation in plant breeding (Merrick et al., 2022). The size of the TRS affects the accuracy of GP models (Goddard, 2009; Daetwyler et al., 2010; Combs and Bernardo, 2013; Bassi et al., 2016) and often correlates positively with the increase in size (Lorenzana and Bernardo, 2009; Zhong et al., 2009; Albrecht et al., 2011; Bentley et al., 2014; Isidro et al., 2015). However, research has shown a plateau in prediction-accuracy increment after reaching an optimum TRS size (Arruda et al., 2015; Sverrisdóttir

et al., 2018; Fernández-González et al., 2023). Increasing the size of TRS demands greater effort and higher costs required for phenotyping as the genotyping cost has been significantly reduced. In addition, increasing the TRS could adversely affect the quality of collected phenotypic data, leading to reduced prediction accuracy. The TRS optimization encircles balancing to achieve the highest possible r_{MG} with minimum resource allocation through selective phenotyping (Figure 3A) (Lorenz and Nice, 2017; Akdemir and Isidro-Sánchez, 2019). Research has been conducted to identify an optimized TRS size and demonstrate the effects of numerous determining factors, such as the genetic kinship and population structure with the BS, LD extent, heritability, and genetic architecture of target traits (Isidro et al., 2015; Akdemir and Isidro-Sánchez, 2019; Sarinelli et al., 2019). Broadly, to achieve a higher r_{MG} , the size of TRS should increase when the genetic kinship with the BS decreases. Likewise, accuracy is often low for less-heritable traits, which is directly related to the complexity of the genetic architecture with several contributing small-effect QTLs and when LD between markers and QTLs is low (Habier et al., 2007; Daetwyler et al., 2010; Clark et al., 2012; Combs and Bernardo, 2013; Wientjes et al., 2013; Isidro et al., 2015). New optimization methods with the capability to automatically

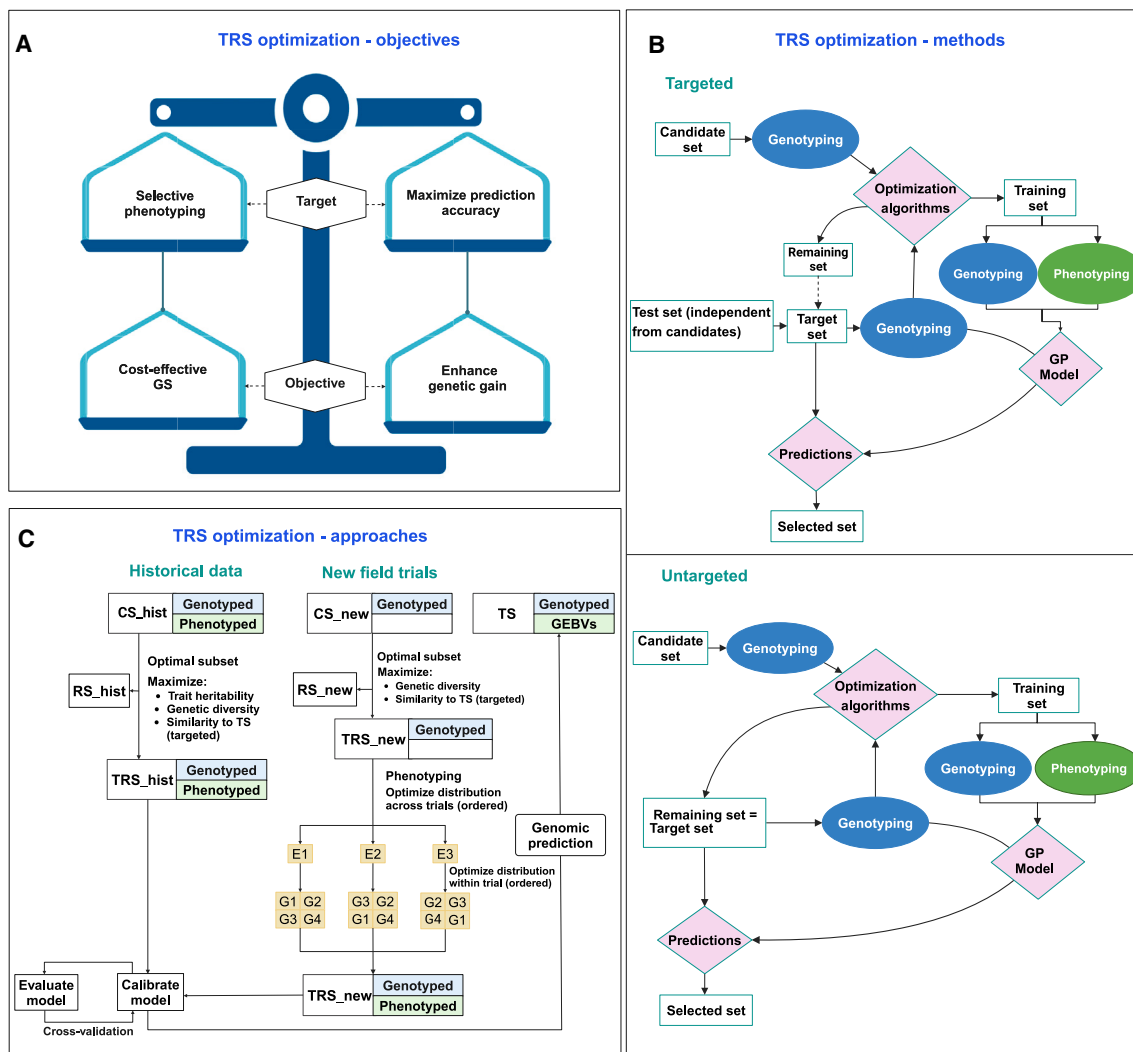


Figure 3. Training population optimization.

(A) The objective of training population optimization is to maximize the genetic gain of plant breeding by enhancing the GP accuracy while minimizing the phenotyping costs by reducing the size of training population.

(B and C) **(B)** The targeted and untargeted training set optimization methods and **(C)** optimization approaches with historical and new field trials data. The aim of TRS optimization is to find a subset of a CS to be used as an optimal TRS to make predictions on a target population of interest. In targeted optimization, there must be a test set containing genotypes different to those in the CS, which is common when working with historical data. The target population includes this independent test set, but it may contain the genotypes in the RS as well if predicting their genotypic values is of interest. The genotypic information of the target population can then be used as an input on the optimization algorithm that allows the abstention of a training set specifically tailored for it. Conversely, in the untargeted TRS optimization scenario, there is no independent test set, which is common for selective phenotyping of new field trials. In this scenario, the target population comprises all genotypes in the CS whose genotypic values are of special interest. The target population would often be equal to the RS, but it can also be the entirety of the CS. The TRS data may be of two types: historical data and data from new field trials. When both data sources are available, their subsequent TRS can be combined to maximize model performance. It is worth noting that the steps highlighted as targeted mandate the availability of genotypic information from the test set. There may also be instances of population overlap within the process should the GEBVs of the remaining set form a prediction target; for example, RS inherently forms a part of the test set. TRS, training set; CS, candidate set; RS, remaining set; TS, test/target set; E, environment; G, genotype.

find the optimal TRS size have been recently developed (Fernández-González et al., 2023, 2024; Wu et al., 2023). More details are available in Supplemental File 1.

Population structure and genetic relationship with breeding population

One of the pitfalls of GP in a practical breeding scenario is the inability to develop a dependent and effective TRS in the long

term without targeting any specific BS. Because of this, breeding programs have to update and optimize the TRS at every single stage where selection is assisted with GP models (see “training population optimization”). This is because the genetic kinship, population structure, and the extent of LD between the training and breeding populations play a huge role in the accuracy. Hence, developing a TRS targeting the candidates for selection is the most critical step in GP (Akdemir et al., 2015; Lorenz and Smith, 2015; Akdemir and Isidro-Sánchez, 2019). Adding

genetically unrelated individuals in the training population adversely affects the GP models, as has been shown with a reduction in r_{MG} (Habier et al., 2010; Clark et al., 2012; Lorenz and Smith, 2015; Alemu et al., 2023). For instance, Riedelsheimer et al. (2013) reported a huge decline (42%) in prediction accuracy when the training and breeding population was changed from within full-sib double haploid (DH) maize lines to between half-sib DH lines.

A specific population having distinct allele frequency from others due to founder effects and selection processes creates population structure (Isidro et al., 2015; Norman et al., 2018). This allele frequency difference often brings association between phenotypic performances with markers, irrespective of their true linkage to the causative QTL, which causes bias on r_{MG} unless properly accounted in the GP statistical machine-learning models (Windhausen et al., 2012; Wray et al., 2013; Albrecht et al., 2014; Guo et al., 2014). In GP, population structure can arise within the TRS or between the TRS and BS, and both affect prediction models. Research indicates an adverse impact of population structure on r_{MG} in both self- and cross-pollinated crops (Windhausen et al., 2012; Riedelsheimer et al., 2013; Hickey et al., 2014; Isidro et al., 2015; Würschum et al., 2017; Werner et al., 2020). However, de Los Campos et al. (2015) argued that natural and artificial breeding populations always have different degrees of stratification due to differences in allele frequency and LD patterns that act as a modifier effect rather than a confounding effect. Daetwyler et al. (2012) mentioned that the key is accounting for spurious population structure, such as that originating from admixtures, but without affecting relatedness between individuals. Nevertheless, several research studies indicated a significant reduction in GP accuracy when population structure was accounted in the statistical analysis (Guo et al., 2014; Norman et al., 2018; Werner et al., 2020; Callister et al., 2022). Different strategies have been proposed to account population structure in GP. Admixing individuals from different groups during TRS optimization and phenotyping is one option to connect the different populations (Esfandiyari et al., 2015; Rio et al., 2019). Accounting population structure by exploiting the mean performances of subpopulations defined through breeding origin, pedigree, or molecular markers is the other developed method (Albrecht et al., 2011; Windhausen et al., 2012; Guo et al., 2014). Another approach is incorporating principal components and admixture coefficients derived from a genomic relationship matrix as covariates in GP mixed models as fixed effects (Daetwyler et al., 2012; Crossa et al., 2016b; Edriss et al., 2017). However, this method has limitations, such as inability to account markers' effect difference across subpopulations (Lehermeier et al., 2015) and "double counting" of population structure (Janss et al., 2012). Different approaches have been proposed to overcome this problem, such as genomic best linear unbiased prediction (G-BLUP) re-parameterization and modeling genetic covariances between individuals from different groups by adapting multi-trait models (Janss et al., 2012; Guo et al., 2014; Lehermeier et al., 2015).

Genetic diversity

Genetic diversity of the TRS is the other major contributing factor in GP (Habier et al., 2007; Lorenzana and Bernardo, 2009; Norman

et al., 2018; Berro et al., 2019). Including individuals with diverse genetic backgrounds helps to capture the full spectrum of genetic variants influencing the target traits. This diversity ensures that the predictive models can accurately capture the genetic effects and make reliable predictions across a wide range of genetic backgrounds. The TRS should encompass a broad range of allelic variation for the traits of interest to capture maximum possible contributing QTLs (Norman et al., 2018). However, it has to be developed targeting the BS, since increasing the diversity with individuals genetically distant from the BS negatively affect the GP model accuracy (Crossa et al., 2014; Akdemir and Isidro-Sánchez, 2019; Berro et al., 2019).

TRAINING POPULATION OPTIMIZATION

The GP efficiency in practical breeding scenarios is highly dependent on the r_{MG} of the genetic merit of candidate individuals. Extensive research supports the notion that configuring the optimal TRS is critical to determine the prediction accuracy (Lorenzana and Bernardo, 2009; Riedelsheimer et al., 2012; Isidro et al., 2015; Akdemir and Isidro-Sánchez, 2019; Berro et al., 2019; Ou and Liao, 2019; Isidro y Sánchez and Akdemir, 2021; Fernández-González et al., 2023). An inadequately constructed TRS substantially diminishes prediction accuracies, while optimized TRS significantly improves accuracy (see Isidro y Sánchez and Akdemir, 2021). The TRS optimization aims to maximize the accuracy of the predictions made on a test or target set (TS) while minimizing the TRS size to reduce phenotyping costs (Figure 3A) (Crossa et al., 2017).

The TRS optimization is key in plant-breeding programs for three main reasons. First, as predictions rely on markers or line effects determined by the TRS, there is a need to carefully curate the TRS to enhance the efficiency and efficacy of GS. Second, the substantial costs of phenotyping have driven the search for innovative alternatives to reduce expenditure (Isidro y Sánchez and Akdemir, 2021). Breeding programs can allocate resources more efficiently by focusing on a smaller yet representative TRS. This not only reduces phenotyping expenditure but also enhances the quality of data applied in the GP models. This allows breeding programs to invest in advanced tools for intricate traits or increase the number of measurements for specific traits, an approach termed sparse or selective phenotyping. Third, the conventional TRS methods that rely on random sampling do not always lead to improved predictive capability due to an under-representation or over-representation of critical genetic information. Thus, optimization serves to streamline the sparse phenotyping process, aiming to curtail phenotyping expenses while preserving or enhancing prediction models' accuracy.

There are two key aspects in TRS optimization: (1) TRS is a dynamic population that must be updated through the breeding cycle (Lorenz and Smith, 2015; Pszczola and Calus, 2016; Akdemir and Isidro-Sánchez, 2019); and (2) the test set needs to be taken into account when building the TRS (Akdemir and Isidro-Sánchez, 2019; Isidro y Sánchez and Akdemir, 2021; Fernández-González et al., 2023).

Here, we review the types of populations available in breeding programs and their role during TRS optimization, the applied

methodologies, and the broader implications on GP accuracy and efficiency. We offer the perspectives of TRS optimization in the context of the broader breeding landscape. We do not delve into the exhaustive details of every algorithm or methods and associated pros and cons that can be found elsewhere, such as [Isidro y Sánchez and Akdemir \(2021\)](#). Nevertheless, a summary of the key developed algorithms for TRS optimization can be found in [Supplemental Table 4](#).

Breeding population types involved in optimization

In GS-assisted breeding, the classification and utilization of different breeding population sets are crucial in streamlining the prediction process and maximizing the efficiency of the breeding pipeline. Each set plays a distinct role, and its composition can significantly influence the accuracy and effectiveness of GP. The summary of breeding population sets and their respective purposes and interrelations can be summarized as follows.

- (1) Candidate set (CS): collection of genotypes available to breeders. Optimization aims to identify an optimal CS subset to be used as the TRS ([Figure 3B](#)).
- (2) Remaining set (RS): includes genotypes from the CS not selected for the TRS. When accompanied by phenotypic data, RS enhances the evaluation of model performance.
- (3) Training or calibration set (TRS): basis for the GP equation, containing both genotypic and phenotypic data. The goal is to maximize accuracy on the TS with minimal phenotypic and genotypic information.
- (4) Test or target set (TS): a set of genotypes to be predicted. It holds only the genotypic information required to predict their GEBVs. However, genotypic information may or may not be available in time for the TRS optimization step.

Optimization scenarios

The TRS is often constructed with new field trials datasets. However, it can be supplemented with old historical data, and optimization can be performed on both data sources ([Figure 3C](#)).

- (1) (Historical data: utilizing a CS that encompasses comprehensive historical data with both genotyped and phenotyped information can enrich the TRS in terms of size and diversity, a key advantage in GS ([Pszczola et al., 2012](#); [Rincet et al., 2012](#); [Isidro y Sánchez and Akdemir, 2021](#); [Fernández-González et al., 2024](#)). Increasing sample size improves the potential to capture the majority of many allele effects and enhance the robustness and accuracy of GP models ([Akdemir and Isidro-Sánchez, 2019](#); [Isidro y Sánchez and Akdemir, 2021](#); [Fernández-González et al., 2024](#)). Such inclusion could however diminish the TRS's resemblance to the TS and may adversely affect the prediction accuracy ([Lorenz and Smith, 2015](#)), prompting the need for optimization.
- (2) New field trials: sparse testing is suggested in cases where the CS provides only genotypic data with limited field trials preventing complete phenotyping ([Crespo-Herrera et al., 2021](#); [Montesinos-López et al., 2023a](#); [Melchinger et al., 2023](#)). In this scenario, an optimal experimental design could be designed as follows: (1) determine the subset from CS to undergo field-testing, thereby forming the

TRS (TRS optimization); (2) for multi-environment trials, ascertain the ideal TRS genotype distribution across locations; and (3) define the most effective genotype distribution within the field (which genotype in which plots). Steps 2 and 3 represent ordered optimization focusing on the strategic optimal spatial arrangement of genotypes.

TRS optimization is categorized as either targeted or untargeted depending on the availability of genotypic information from the TS ([Figure 3B](#)). Targeted optimization takes advantage of TS genotypic information to construct the TRS and often outperforms the untargeted approaches ([Akdemir and Isidro-Sánchez, 2019](#); [Isidro y Sánchez and Akdemir, 2021](#); [Fernández-González et al., 2023](#)). Even without TS genotype information, a detailed pedigree linking the CS with TS remains feasible in targeted optimization. However, research in this area is lacking. Unordered optimization focuses on selecting a CS subset, while ordered optimization emphasizes the spatial genotype distribution in the field. The latter may utilize data related to blocking structures, spatial influences, and environmental variables ([Akdemir et al., 2021](#)).

Training population optimization algorithms

Several design criteria have been proposed for selecting and optimizing the TRS in GP. The classical standard random or stratified sampling method is commonly applied because of its simplicity. Nevertheless, GP accuracy enhancement has been achieved using other optimization criteria, which can be classified as parametric, non-parametric, and multiple design criteria. Many of the established criteria mostly serve as evaluation metrics for the TRS, and appropriate heuristic is imperative to maximize or minimize it. Numerous R program packages have been developed and provide suitable heuristics often based on genetic algorithms. For instance, the STPGA ([Akdemir, 2017](#)), TSDFGS ([Ou and Liao, 2019](#)), and odw ([Butler et al., 2013](#)) are developed but are limited to built-in criteria. In contrast, TrainSel ([Akdemir et al., 2021](#)) supports both built-in and user-defined criteria.

Parametric design criteria

Parametric design criteria assume that the researcher predetermines a model prior to data collection. These criteria typically rely on a scalar function tied to the model's information matrix. In practice, it is usually derived from the prediction error variance-covariance matrix (PEV) for the additive genotypic effects in linear mixed models. The A, D, and E criteria ([Laloë, 1993](#)), the coefficient of determination criterion (CD_mean), and the prediction error variance criterion (PEV_mean) are examples ([Laloë, 1993](#)). Parametric criteria are a powerful approach but are computationally intensive. Attempts have been made to resolve this problem, including updating the PEV matrix in each iteration instead of calculating *de novo* ([Butler et al., 2013](#)) and applying principal component analysis to reduce dimensionality such as in PEV_mean^{ridge} and CD_mean^{ridge} methods ([Akdemir et al., 2015](#); [Heslot and Feoktistov, 2020](#)). An in-depth discussion on computational efficiency of algorithms is available in [Supplemental File 1, Note 2](#).

Sparse selection index is a recently proposed prediction model with a built-in optimization process ([Lopez-Cruz and de los Campos, 2021](#); [Lopez-Cruz et al., 2021](#); [Lopez-Cruz et al.,](#)

2022). Here, a selection index that specifies the TS genotypic values as a linear combination of the CS ones is defined. The regression coefficients of the linear combination are subjected to a lasso regularization (L1) penalty to enforce sparsity that is equivalent to the selection of a subset of the genotypes as a TRS. This is conceptually similar to the bandwidth parameter in the reproducing kernel Hilbert space (De Los Campos et al., 2009; Lopez-Cruz et al., 2021) but takes it one-step further. This method is suitable for historical data optimization because it makes a specific TRS for each TS individual and the phenotypic information of the CS should be available for parameter tuning.

Non-parametric design criteria

The methods of this type do not assume any predefined underlying models but often revolve around metrics of distance or similarity with the intention of uniformly distributing the TRS throughout the design landscape, a method known as space-filling design. Such designs particularly facilitate the selection of a condensed set of candidates and minimize the computational complexity associated with the optimization of parametric design criteria. Various metrics assist in evaluating the distribution of points within this design space. For instance, the partition around medoids approach centers on identifying a series of central entities, termed medoids, within clusters based on a specific distance measure (Guo et al., 2019). In general, methods for representative subset selection in data mining could be used for training set design, which opens up many possibilities. Numerous metrics have been developed to minimize genetic relationships within the TRS (i.e., maximizing diversity) and/or maximizing its relationship to the TS, for example, the maximin and minimax (Johnson et al., 1990), Avg_GRM (Atanda et al., 2021a), OPT_MIN (Lemeunier et al., 2022), Avg_GRM self, and Avg_GRM_MinMax (Fernández-González et al., 2023). Latin hypercube sampling (Helton and Davis, 2003) involves segmenting the design space into equal cubes. The objective is to ensure that each cube hosts a single sample point, further aiming to comprehensively explore the range of each scalar input in alignment with a given probability distribution. Tails and Tails_GEBVs select genotypes with extreme phenotype or GEBVs for the TRS and discard the rest (Neyhart et al., 2017; Fernández-González et al., 2024). Adversarial selection tries to ensure that the TRS and TS are indistinguishable by a binary classifier; i.e., their similarity is maximized (Montesinos-López and Montesinos-López, 2023).

Multi-objective criteria

This method attempts to handle the choice issue by combining the different criteria into one with some type of averaging methods such as the Pareto front approach (Akdemir et al., 2015; Isidro y Sánchez and Akdemir, 2021). It is adept at evaluating multiple criteria and defining a suite of non-dominated designs. The method has been effectively applied to optimize the integration of historical data balancing the TRS diversity, its association with the TS, and trial heritability with an extensive empirical dataset from an industrial breeding program (Isidro y Sánchez and Akdemir, 2021; Fernández-González et al., 2024).

Summary of the key algorithms

A detailed comparison and breakdown of TRS optimization methods and algorithms can be found in Supplemental Table 4. However, the large number of available methods makes selecting a single effective method challenging. Recent research on TRS optimization offers invaluable insights into selection

of appropriate methods. Fernández-González et al. (2023) conducted an extensive comparison of these optimization methods across various datasets and genetic architectures. In light of their findings and those of other researchers, we provided a small summary focusing on the key algorithms for each field within TRS optimization. Furthermore, an in-depth, systematic example on the implementation of CDmean and Avg_GRM_self is provided in Supplemental File 1, Note 1 and an R-script with example of implementing two algorithms on real TRS optimization scenario (Supplemental File 2). Nevertheless, it is important to note that there is no single best algorithm in all aspects, and some of the methods in Supplemental Table 4 may be more suitable for niche applications.

Here, we forward our recommendation of general-purpose, effective methods that constitute a good first option for new optimization projects.

- (1) TRS size optimization: tailored mainly for situations incorporating historical data, two main algorithms emerge:
 - Target accuracy methods: aimed at predicting GP accuracy and identifying the minimal TRS size without substantial accuracy loss (Fernández-González et al., 2023; Wu et al., 2023). In our experience Avg_GRM_self is the best option due to its fast computational time, essential in this application. Importantly, budgetary constraints play a crucial role, but typically, including 50%–85% of the entire candidate population maintains an accuracy decrease below 5%.
 - Best solution methods: these seek the optimal TRS size by identifying local maxima or inflection points. For example, Avg_GRM_MinMax (Fernández-González et al., 2023) and Min_GRM (Fernández-González et al., 2024).
- (2) Optimizing TRS composition: an extensively researched area. Findings suggest that targeted optimization usually surpasses untargeted methods, with CDmean being highly efficient, albeit computationally intensive. Maintaining TRS diversity is especially important in the presence of a strong population structure. Therefore, it is advisable to apply CDmean for smaller datasets, while the fast Avg_GRM_self (untargeted) or Avg_GRM_MinMax (targeted) are suitable for larger datasets.
- (3) Simultaneous size and composition optimization: beneficial when utilizing historical data and the training set size is not determined by available field resources, although it may reduce the versatility of the algorithm for optimization of new field trials due to the potential difficulty of matching optimal TRS size to actual field resources. MaxCD (Guo et al., 2019) was originally described for TRS design in hybrid breeding, but we believe that its ability of optimizing TRS size could be useful for optimization in historical data. The latter role could be filled by other methods such as adversarial selection (Montesinos-Lopez et al., 2023a, 2023b) or multi-objective optimization (Akdemir et al., 2021; Fernández-González et al., 2024), which are not specific to hybrids.
- (4) Spatial distribution/ordered optimization: pertinent for new field trials, this optimization is computationally demanding, especially when incorporating environmental or spatial data. Two notable R package algorithms, “odw” (Butler

et al., 2013) and “TrainSel” (Akdemir et al., 2021), have been developed for this purpose. Parametric criteria such as A-opt and CDmean are the best-suited approaches for this application

DENSITY AND DISTRIBUTION OF MARKERS AND LINKAGE DISEQUILIBRIUM

Increasing the density of SNP markers distributed across chromosomes helps to accurately capture most contributing QTL ultimately leading to an increased r_{MG} . The number of SNP markers required to develop an optimum GP depends on the genome size, extent of LD, and complexity of the trait under investigation. A study targeting a complex trait controlled by several QTLs (e.g., yield) in a crop with a large genome size and low LD relatively requires a highly dense SNP marker distributed across chromosomes. On the contrary, a highly heritable trait controlled by fewer genes and with high LD could need a relatively low SNP marker density to reach the maximum possible r_{MG} . In addition, LD in outcrossing crops, such as maize, decays rapidly compared to self-crossing crops (e.g., rice) (Flint-Garcia et al., 2003; Kaler et al., 2022), requiring highly dense SNP marker distribution to achieve the optimum r_{MG} . In general, the optimum density and distribution of SNP markers relies on the most contributing QTL of a target trait being under LD with DNA markers included in prediction models (Hayes and Goddard, 2001; Kaler et al., 2022). The pattern of LD of populations particularly helps to develop GP models with cost-effective, low-density SNP markers (Bolormaa et al., 2015; Wu et al., 2016; Silva et al., 2018; Ballesta et al., 2020).

Optimizing the marker density in GP could be beneficial, as most SNPs in large marker datasets are phenotypically neutral and contain only a relatively small proportion of SNPs relevant for a specific trait (Bermingham et al., 2015; Al Kalaladeh et al., 2019; Weber et al., 2023). Selecting optimal subsets of markers for specific traits has been a promising approach to increase the accuracy in GP (Bermingham et al., 2015; van den Berg et al., 2016; Filho et al., 2019; Alemu et al., 2023; Weber et al., 2023). One method for marker subsetting is selection based on previous association mapping studies. A beneficial GP accuracy improvement was observed when significant markers identified through GWAS were fitted as fixed effects (Kim et al., 2022; Anilkumar et al., 2023; Chen et al., 2023), only the top 100–10 000 markers with highest significance were used as predictors (Bermingham et al., 2015; Filho et al., 2019), or markers surrounding the significant markers were included (van den Berg et al., 2016; Filho et al., 2019). Another optimization approach is applying haplotype blocks based on marker LD in prediction models (Alemu et al., 2023; Weber et al., 2023). Predictions based on haplotype blocks, rather than single SNP markers, could efficiently capture local epistasis and better account for LD to QTLs leading to improved GP accuracy (Weber et al., 2023). Selection of marker panels can highly influence trait associations, and several research articles have demonstrated the impact of marker density on the GP accuracy (Zhang et al., 2017a, 2019; Liu et al., 2018; Norman et al., 2018).

GENETIC ARCHITECTURE AND HERITABILITY OF TRAITS

Most crop traits of economic importance, such as yield, are multi-genic and have a complex genetic architecture involving several QTLs or genes with varying levels of phenotypic effects. One of the key advantages of GP over the conventional MAS methods is its ability to efficiently evaluate genotypes for such genetically complex multigenic traits by considering the high numbers of small-effect QTLs. Generally, genetic complexity and heritability (h^2) are directly related to the number of QTLs and their interaction to control a trait. A trait controlled by small numbers of large-effect QTLs usually has higher heritability than those with several genes with different levels of genotypic effect. GP is affected by the complexity of traits, genetic architecture, and heritability. A trait with low h^2 should be compensated by increasing the TRS size (N) to achieve an optimum GP accuracy, since Nh^2 determines the power of GP models (Bernardo, 2016). Furthermore, machine-learning models that account for epistatic interactions have the potential to improve the prediction accuracy when epistatic interaction largely or partially contributes to the true genetic architecture of a trait (De Los Campos et al., 2010; Wang et al., 2012; Morgante et al., 2018). Several empirical investigations and simulation research have demonstrated that the r_{MG} generally increases as the number of QTLs decreases and trait heritability increases (Hayes et al., 2009; Lorenzana and Bernardo, 2009; Zhong et al., 2009; Jannink et al., 2010; Combs and Bernardo, 2013; Zhang et al., 2017a; Jung et al., 2020).

PRECISION PHENOTYPING

The phenotypic data recorded from the TRS is required to connect the genomic profile with the phenotype, enabling GP models to evaluate and provide weights to individual SNP markers. These markers are then used to assess individuals in the BS solely from their genomic profile and assist selection and decision making in breeding programs. High-density SNP markers combined with precision phenotyping evaluated in suitable statistical machine-learning models could link the genome with the phenome of crops, leading to GP models with high prediction accuracy. Efficiency limitations of the conventional plant phenotyping methods have been considered as the bottleneck to successfully connecting the bridge between genotype with phenotype information (Araus and Cairns, 2014; Araus et al., 2018). Hence, advanced technologies for high-throughput phenotyping (HTP) and high-throughput field phenotyping (HTFP) methods have attracted tremendous attention recently for their potential to provide comprehensive and precise phenotypic data for primary as well as secondary traits in several crops (Cabrera-Bosquet et al., 2012; Araus and Cairns, 2014; Zhang et al., 2017b; Araus et al., 2018; Moreira et al., 2020). The HTP and HTFP can be referred to collectively as the high-throughput phenotyping platform (HTPP). The HTPP allows researchers to screen massive numbers of individual plants at a very low cost. HTPP aims to produce high-density phenotypes on very large numbers of individuals or breeding lines across time and space at low cost using remote or proximal sensing. This can increase both the accuracy and intensity of selection and, therefore, the selection response while decreasing phenotyping costs. The main idea of HTPP is to use predictor traits related to grain yield, disease

resistance, or end-use quality that could be advantageous in early-generation testing of lines (Rutkoski et al., 2016). Previous research has shown the potential of HTPP methods in the GP accuracy of several traits (Crain et al., 2018; Juliana et al., 2019a; Galán et al., 2020; Wang et al., 2023b).

INTEGRATING OTHER OMICS DATA

GP relies on estimating the phenotypic performance of individuals from their genomic profile. The genomic profile, however, must be transcribed to RNA (tRNA, sRNA, mRNA) and then translated to protein before being expressed as a phenotype (Cobb, 2017). These results come from transcriptomics and proteomics research, respectively. The integration of this intermediate phenotype data (i.e., transcriptomics, proteomics, and metabolomics) with genomic data has demonstrated the potential to improve GP accuracy (Hu et al., 2019; Li et al., 2019; Haile et al., 2020; Martini et al., 2022; Wang et al., 2023a, 2023b). Multi-omics-based GP has been implemented successfully, improving prediction accuracy in several crops such as maize (Guo et al., 2016; Zenke-Philippi et al., 2016; Westhues et al., 2017; Xu et al., 2017; Schrag et al., 2018), wheat (Zhao et al., 2015), oats (Hu et al., 2021), barley (Wu et al., 2022), rice (Hu et al., 2019; Wang et al., 2019), and rapeseed (Knoch et al., 2021).

Schrag et al. (2018) reported combining messenger RNA (mRNA) with pedigree and genomic datasets, resulting in beneficial improvements in r_{MG} to estimate the breeding values of agronomic traits in untested maize hybrids. Hu et al. (2019) outlined an r_{MG} improvement in four yield and yield-related traits of untested rice RILs through a multi-layered least absolute shrinkage and selection operator model integrating transcriptome and metabolome along with genomic profiles in a single model. Incorporating both transcriptomic and metabolomic profiles to the genomic datasets has also improved the r_{MG} of several agronomic and seed nutritional traits of oats from multi-environment trials (Hu et al., 2021). Recently, advanced statistical machine-learning algorithms have been developed to incorporate the multi-omics intermediaries with efficient computing performance to leverage the GP models (Hu et al., 2021; Wang et al., 2023a, 2023b). Nonetheless, model overfitting and spatial-temporal features accompanying the intermediaries should be cautiously considered during implementation of the omics profiles of plants in GP research (Yan and Wang, 2023).

STATISTICAL MACHINE-LEARNING METHODS

Statistical methods play a central role in GP, since the effect of DNA markers is estimated by modeling the mathematical relationships between the genotypic and phenotypic data provided in the TRS. Thereafter, evaluated markers are provided with specific weights to their phenotypic effect that allow the genomic breeding values of candidate individuals in the BS to be estimated. Hence, GP is a statistical machine-learning approach that aims to train, develop, and analyze the performance of models with the data from the TRS (Tong and Nikoloski, 2021; Montesinos López et al., 2022a, 2022b, 2023). Meuwissen et al. (2001) by simulating the effect of approximately 50 000 marker haplotypes with a modified linear least squares regression,

BLUP, and two Bayesian statistical methods (BayesA and BayesB).

Several statistical machine-learning methods have been proposed for GP over the last 20 years (Figure 4). As most of the available datasets in genomics for plant and animal breeding applications try to find the relationship between the response variable (output) and thousands or even millions of SNP markers as inputs (or predictors, p), the framework for training these models is where more inputs are available than observations (observations, n), that is, $p \gg n$, which presents a major challenge. This phenomenon leads the advent of different penalization (regularization) approaches (Meuwissen et al., 2001; De Los Campos et al., 2013). Hence, the different regularization mechanisms applied to estimate genome-wide SNP marker effects from a regression with large p with small n has led to the emergence of numerous statistical machine-learning approaches (Montesinos López et al., 2022a, 2022b). These statistical machine-learning algorithms perform differently, and their suitability and performance depend on coping with several factors that arise from the multi-dimensionality of genome-wide SNP markers and the genetic complexity of multi-factorial traits (De Los Campos et al., 2010). Consequently, no statistical machine-learning model can be singled out as outperforming other available algorithms and giving the highest possible GP accuracy that has been verified by numerous empirical and simulation researches and its theoretical support given by the “no-free-lunch” theorem (Azodi et al., 2019; Montesinos López et al., 2022a, 2022b). These statistical models can be grouped as parametric, semi-parametric, and non-parametric models (Montesinos López et al., 2022a, 2022b).

Parametric models are developed from the assumption that the independent or predictor variables take a predetermined function with the dependent or response variables. Some examples of parametric models are linear multiple regression, non-linear regression, logistic regression, multinomial regression, and Poisson regression (Montesinos López et al., 2022a, 2022b). Non-parametric models are a class of statistical and machine-learning models that do not make explicit assumptions about the functional form or distribution of the underlying data. Predictors are not predefined in this class of models but are instead crafted on the basis of insights extracted from the data (Montesinos López et al., 2022a, 2022b). Unlike parametric models, which assume specific mathematical forms for relationships between variables (e.g., linear regression), non-parametric models offer more flexibility by allowing the data to determine the structure of the model. These models are particularly useful when dealing with complex or unknown relationships, as they can adapt to various data patterns without requiring predefined parameter specifications. Non-parametric models include methods such as kernel density estimation, k -nearest neighbors, decision trees, gradient boosting machine, and random forest. A semi-parametric model is a statistical machine-learning approach where a portion of the predictors is not constrained to predetermined mathematical forms, while another portion adheres to known functional relationships with the response variable. This blend of flexibility and structure is exemplified by equations such as

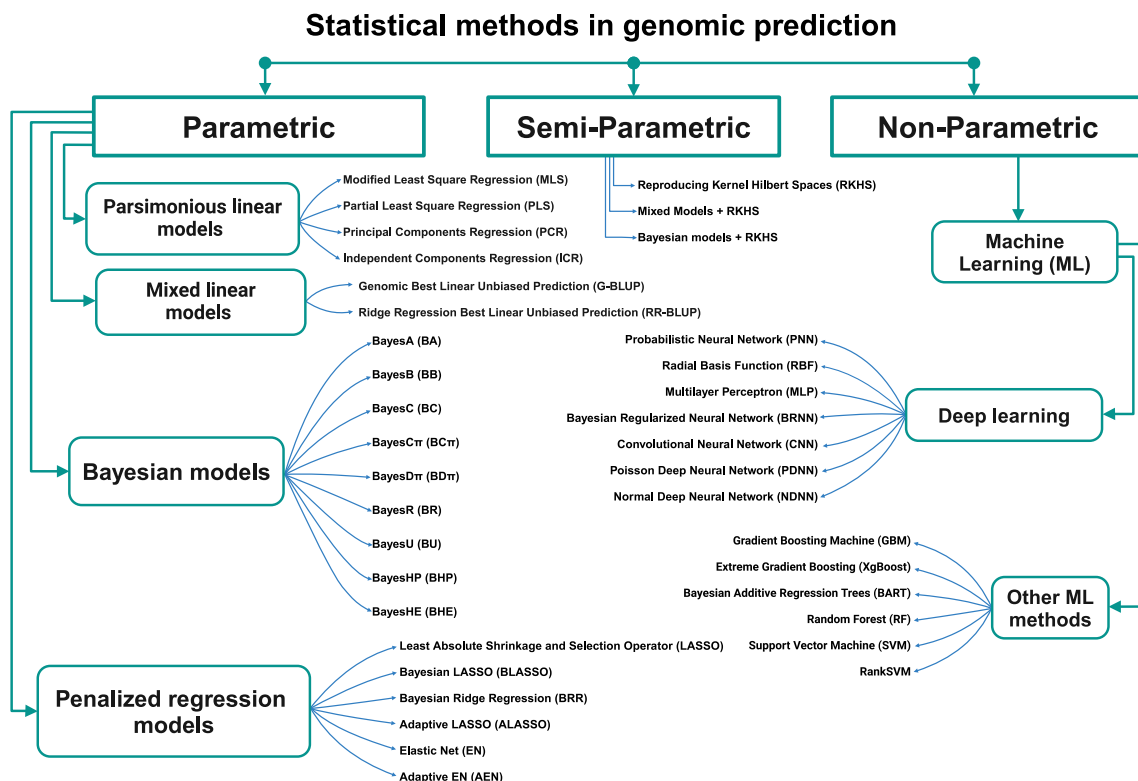


Figure 4. List of the statistical machine-learning models currently in use for genomic prediction.

All these statistical machine-learning models are classified into three major categories: parametric, semi-parametric, and non-parametric. The parametric statistical machine-learning models include modified least square regression (MLS) (Meuwissen et al., 2001), partial least square regression (PLS) (Montesinos López et al., 2022a; 2022b), principal components regression (PCR) (Solberg et al., 2009), independent components regression (ICR) (Azevedo et al., 2013), genomic best linear unbiased prediction (G-BLUP) (Vanraden, 2008), BayesA (BA) and BayesB (BB) (Meuwissen et al., 2001), BayesC (BC) (George and McCulloch, 1993), BayesC π (BC π) and BayesD π (BD π) (Habier et al., 2011), BayesR (BR) (Erbe et al., 2012), BayesU (BU) (Pong-Wong and Woolliams, 2014), BayesHP (BHP) and BayesHE (BHE) (Shi et al., 2021), least absolute shrinkage and selection operator (LASSO) (Usai et al., 2009), adaptive LASSO (ALASSO) (Zou, 2006), Bayesian LASSO (BLASSO) (Park and Casella, 2008), ridge regression best linear unbiased prediction (RR-BLUP) (Meuwissen et al., 2001), Bayesian ridge regression (BRR) (Pérez et al., 2010), elastic net (EN) (Zou and Hastie, 2005), and adaptive EN (AEN) (Zou and Zhang, 2009). The semi-parametric method includes the reproducing kernel Hilbert space (RKHS) model (Gianola et al., 2006) and the mixed and Bayesian models combined with the RKHS model. The non-parametric method comprises gradient boosting machine (GBM) (Li et al., 2018), extreme gradient boosting (XgBoost) (Chen and He, 2014), support vector machine (SVM) (Maenhout et al., 2007), rankSVM (Blondel et al., 2015), Bayesian additive regression trees (BART) (Waldmann, 2016), random forest (RF) (Chen and Ishwaran, 2012), probabilistic neural network (PNN) (González-Camacho et al., 2016), radial basis function (RBF) (Chen and Ishwaran, 2012), multilayer perceptron (MLP) (Gianola et al., 2011), Bayesian regularized neural network (BRNN) (Pérez-Rodríguez et al., 2012), convolutional neural network (CNN) (Ma et al., 2018), Poisson deep neural network (PDNN), and normal deep neural network (NDNN) (Montesinos-López et al., 2020).

$$y = \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + m(x) + \epsilon, \quad (\text{Equation 1})$$

$$y = \exp(\beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3) + m(x) + \epsilon. \quad (\text{Equation 2})$$

In the context of GP models, a classical example is Bayesian or mixed models with linear components for environmental effects and non-linear (Gaussian kernel, or other types of kernels) components for genotype effects (Montesinos López et al., 2022a, 2022b). Essentially, semi-parametric models represent a combination of both parametric and non-parametric modeling techniques. Most of the currently available statistical machine-learning models classified in the three mentioned groups are presented in Figure 4.

Modeling genotype $\times$ environment interaction

In studies involving multiple environments, genetic association and prediction models are usually developed from summarized

phenotypic data across environments or separate models for each environment. Another approach to account for multiple environments is using an environment index, which for example can be derived from environmental conditions such as temperature and photoperiod (Guo et al., 2020a, 2020b, 2020c, 2020d). Growing degree days was earlier proposed as a promising example of an environment index for capturing flowering time plasticity in rice (Guo et al. 2020a). Li et al. (2021) proposed that a carefully developed environment index can replace phenotypic means obtained through conventional measurements and can model observed phenotype and also predict phenotypic performance in new environments, and they tested their hypothesis on three different traits in wheat and oat field trials. Similarly, in sorghum, diurnal temperature range during the rapid growth period was found to be an effective environment index (Mu et al., 2022). Taken together, these studies highlight the importance of studying phenotypic

plasticity under G×E interaction and exploring derived environmental indices for modeling and predicting phenotypes in untested environments.

The reaction norm model

Multi-environment trials for assessing G×E play an important role in plant breeding for selecting high-performing and stable lines across environments. For instance, the multi-environment linear mixed models accounting for correlated environmental structures within the G-BLUP framework increased accuracy when predicting the performance of unobserved phenotypes using pedigree and molecular markers (Zhang et al., 2015). Burgueño et al. (2012) proposed and effectively applied a marker and pedigree G-BLUP model for assessing G×E, while Heslot et al. (2014) incorporated crop modeling data on the genomic G×E prediction. Jarquín et al. (2014) developed a reaction norm model, an extension of the G-BLUP model, where the main and interaction effects of markers and environmental covariates are introduced using highly dimensional random variance-covariance structures of markers and environmental covariables. The model has been successfully applied in GP prediction of breeding values using pedigree and genomic relationships (Pérez-Rodríguez et al., 2015; Velu et al., 2018).

Here, the baseline model for the phenotypes (y_{ij}) can be described as

$$y_{ij} = \mu + E_i + L_j + EL_{ij} + e_{ij}, \quad (\text{Equation 3})$$

where μ is the overall mean, E_i ($i = 1, \dots, I$) is the random effect of the i th environment, L_j is the random effect of the j th line ($j = 1, \dots, J$), EL_{ij} is the interaction between the i th environment and the j th line, and e_{ij} is the random error term. The assumptions are as follows: $E_i \sim N(0, \sigma_E^2)$, $L_j \sim N(0, \sigma_L^2)$, $EL_{ij} \sim N(0, \sigma_{EL}^2)$, and $e_{ij} \sim N(0, \sigma_e^2)$, with $N(\cdot, \cdot)$ denoting a normal density and iid standing for independent and identically distributed. Markers can be introduced in Equation 3 such that the effect of line (L_j) can be replaced by g_j defined by the regression on marker covariates (it approximates the genetic value of the j th line). The vector containing the genomic values is $g \sim N(0, \mathbf{G}\sigma_g^2)$, where σ_g^2 is the genomic variance and $\mathbf{G}$ is a genomic relationship matrix (Vanraden, 2008). Furthermore, the effects of line (L_j) can be replaced by a_j , with $a \sim N(0, \mathbf{A}\sigma_a^2)$, where $\mathbf{A}$ is the additive relationship matrix derived from pedigree and σ_a^2 is the additive variance.

The marker × environment interaction model

The marker × environment (M×E) interaction model proposed by Lopez-Cruz et al. (2015) breaks down the marker effects into components that are common across environments (stability) and environment-specific deviations (interaction). This model borrows information across environments while allowing marker effects to change across environments. This method can be implemented using both shrinkage and variable selection methods and thus can be used to identify genomic regions with stable effect across environments and regions that are responsible for G×E. However, it is noteworthy that the M×E model is best suited for joint analysis of positively correlated environments (Lopez-Cruz et al., 2015). Crossa et al. (2016a) successfully applied the M×E GP model to predict untested individuals and dissect

genomic regions with stable effect across environments and with environment-specific effect.

IMPLEMENTING GENOMIC PREDICTION AT DIFFERENT BREEDING STAGES

There are various uses of GP in breeding crops. The first is in pre-breeding, either to search for desired accessions based on their GEBVs in a gene bank (Crossa et al., 2016b; Dzevit et al., 2021; Bohra et al., 2022; El Hanafi et al., 2023) or to identify elite parents for further crossing (Gaynor et al., 2017; Chung and Liao, 2022). GP allows a cost-effective approach for selecting interesting germplasm held in gene banks (Yu et al., 2016), thus increasing the use of this germplasm—particularly those lacking pedigree information and data evaluation—in plant breeding (Jiang et al., 2021). It also speeds up the introgression of exotic germplasm into the elite breeding pool (Crossa et al., 2016b), as shown recently in wheat improvement (Schulthess et al., 2022). GP may also be used for increasing genetic gains by selecting promising germplasm at early stages (Kadam et al., 2016; Rembe et al., 2022) or for feeding them into a genomic recurrent selection (GRS) approach (Bassi et al., 2016; Biswas et al., 2023), as well as for accelerating the cultivar development pipeline (Ballén-Taborda et al., 2022). GRS facilitates the recycling of parents in a breeding program. The success of GP in any of these breeding stages, however, relies mainly on the trait architecture and its heritability.

A challenge faced by plant breeding is to predict performance across sites over years or cropping seasons. GP may allow estimation of the robustness of desired productivity or quality traits across the target population of environments. Such an approach improves the efficiency of multi-environment testing and its further use in the cultivar development pipeline because it eliminates mediocre breeding lines in the early stages, thus saving time and resources. In this regard, as shown by Atanda et al. (2021b), sparse testing using GP may also be a valuable approach for increasing the number of trial environments without increasing costs but keeping the selection intensity in the early stages of evaluation. Montesinos-López et al. (2023b) showed that a significant gain in the number of new lines to be evaluated could be obtained by using sparse testing methods without a relevant increase of required resources. The authors demonstrated that with a conventional block design capacity to evaluate only 225 lines, the number could be increased to 269, 308, and 475 with a sparse testing design using 85%, 75%, and 50% as training increasing the number of lines by 19.56%, 36.89%, and 111.11%, respectively.

GP has further found extensive application in predicting heterosis, encompassing both high-parent and mid-parent heterosis, across a diverse range of crops, including maize (Albrecht et al., 2011, 2014; Riedelsheimer et al., 2013; Beyene et al., 2015, 2019; Cantelmo et al., 2017; Zhang et al., 2022), rice (Xu et al., 2014, 2018; Huang et al., 2015; Cui et al., 2020), barley (Philipp et al., 2016; Li et al., 2017), wheat (Basnet et al., 2019; Zhao et al., 2021), sorghum (Sapkota et al., 2022; Kent et al., 2023; Maulana et al., 2023), ryegrass (Grinberg et al., 2016), and pumpkin (Wu et al., 2019). Notably, the predictive scope of GP extends beyond conventional traits such as yield and its

components (Grinberg et al., 2016; He et al., 2016; Philipp et al., 2016; Wu et al., 2019) to encompass a wider spectrum of characteristics, such as biotic and abiotic stress tolerances (Lorenz et al., 2012; Aroju et al., 2018), nutrient utilization efficiency (Zhao et al., 2020), and biofortification of crops with several micronutrients (Velu et al., 2016; Mageto et al., 2020; Rakotondramanana et al., 2022; Tadesse et al., 2023).

ACHIEVEMENTS

The task of applying GS in breeding is to enhance genetic gains per year at a lower cost and in less time compared to the conventional breeding methods. Given a vector of true breeding values of an individual $\mathbf{a}' = [a_1 \ a_2 \dots a_t]$ and the vector of economic weights $\mathbf{w}' = [w_1 \ w_2 \dots w_t]$ for t traits, the net genetic merit is $H = \mathbf{w}'\mathbf{a}'$. The response to multi-trait genetic gains can be written as $H = (k\sigma_H\rho_{H,I})/L$, where k is intensity of selection (the standardized selection differential), σ_H is the standard deviation of H , $\rho_{H,I}$ is the correlation between H and any phenotypic or genomic index I , and L is the time required for I to complete one selection cycle (in a standard breeding program this takes several years). The selection response is the most important breeder's equation, and factors that increase the numerator or decrease the denominator of R will increase the overall genetic gains of the target traits. Simulation and empirical results have shown that GS can increase genetic gains by shortening the breeding interval cycle (L) (rapid selection cycle) or increasing testing efficiency by performing sparse field evaluation (Tessema et al., 2020; Xu et al., 2020; Atanda et al., 2022; Dreisigacker et al., 2023). To achieve a shorter interval cycle (I), the most favorable situation for GS is prediction within full-sib families, since the biparental populations have very high LD between marker alleles and QTL alleles with no pedigree, family, or group structure. Estimated prediction accuracies for biparental populations should thus be considered the maximum attainable in closed rapid-cycle marker-only selection. Several research confirmed the efficiency of GS for early-generation rapid cycling (Massman et al., 2013; Zhang et al., 2017c; Bonnett et al., 2022; Dreisigacker et al., 2023).

Two showcases are provided to elucidate the ongoing empirical research facilitated by GS from public and private breeding programs.

Showcase 1: Genetic gains of maize in Africa

Most GS results in maize have been achieved by rapid cycling of biparental populations. For example, the $F_{2:3}$ segregating populations were crossed with a tester, usually from the opposite heterotic group. CIMMYT's Global Maize Program designed a GS rapid cycle of multi-parental crosses. Fifteen elite tropical maize lines were crossed in diallel fashion to form cycle 0 (C_0) comprising 1000 plants, which were genotyped with 1 000 000 genotyping-by-sequencing (GBS) SNP markers and phenotyped at three locations in Mexico. The best phenotypic plants were selected to form the parents for GS cycle 1 (C_1). The C_1 parents were intercrossed and the progeny was genotyped with the same GBS markers as used for the C_0 population. Genomic-enabled prediction for grain yield for the individuals in the C_1 population was performed in each of the three environments; based on the predicted values, selection was made to form the parents of the C_2 population. As before, the parents were intercrossed and genotyped to form the

C_2 population, and plants were selected based on the GP for grain yield. GP and GS were performed for two more cycles. Two cycles per year were performed; and at the end of the second year, seeds from cycles C_0 , C_1 , C_2 , C_3 , and C_4 were collected, assembled, and sown at three locations in Mexico (Agua Fria, Cotaxtla, and Tlaltizapan). Fifty entries were sown per genomic cycle at each location, together with two widely used commercial tropical maize hybrids. The average genetic grain yield gains were 0.134 t ha^{-1} with C_0 producing 6.653 t ha^{-1} . Grain yield of C_1 was slightly lower (6.488), and cycles C_2 , C_3 , and C_4 produced means of 7.022 , 6.879 , and 7.126 t ha^{-1} , respectively. Cycles C_2 and C_4 were significantly different from the rest (least significant difference at the 0.05 probability level). Results from two other locations in Mexico are being processed, and the complete results of this multi-parental maize rapid selection cycle are yet to be published.

In addition, Beyene et al. (2015) previously reported significant genetic gains in maize grain yield through GS in eight CIMMYT tropical biparental maize populations in sub-Saharan Africa under drought conditions. They revealed that the average gain from GS per cycle across the eight populations was 0.086 t ha^{-1} , while the C_3 -derived hybrids produced significantly higher average grain yields than C_0 -derived hybrids. However, the average gain per cycle using marker-assisted recurrent selection across 10 populations was only 0.045 t ha^{-1} per cycle under similar environmental conditions.

Showcase 2: Two-part GS-assisted breeding at Lantmännen Lantbruk, Sweden

The breeding-cycle duration is arguably the single factor that has the largest effect on gain per time (Cobb et al., 2019). The genetic gain per unit time is of fundamental importance, particularly for breeding programs to maintain their competitive advantage, and is also crucial for attempting to adapt new cultivars to a rapidly changing environment (Budhlakoti et al., 2022). In a conventional breeding program of an inbred crop, such as wheat, barley, or oats, new parents are typically selected during the advanced yield trial stage, which results in a breeding cycle of around 5–8 years.

In Lantmännen, the GS-assisted breeding program of inbred crops is split into two parts: the first part is the GS-enabled recurrent selection, also called "population improvement"; and the second part is inbred line development, also called "product development," in which selected lines undergo testing in advanced field trials. This strategy significantly reduces the breeding-cycle time by selecting new parents at an early stage based on their genomic estimated breeding values. Simulation research supports this two-part strategy, outperforming both the conventional selection as well as "standard" GS (i.e., GS only applied at the preliminary yield trial stage) by significantly increasing genetic gain per unit time (Gaynor et al., 2017). Meanwhile, the two-part GS-assisted breeding strategy brings challenging issues for breeding programs. First, genotyping a large number (up to 100 000) of early-generation individuals for high-density SNP markers could be expensive, particularly for small breeding programs. Second, a closed-loop two-part strategy, where no new allelic variation is introduced, leads to loss of both genetic diversity and prediction accuracy over time, with a negative impact on long-term genetic gain (Gaynor et al., 2017).

However, for self-pollinating crops where LD normally extends over longer genomic distances, rapid-cycling GS using a low-density marker set can deliver gains of similar magnitudes as high-density genotyping, even without marker imputation (A. Ceplitis, Lantmännen Lantbruk, Svalöv, Sweden, unpublished data). In addition, the negative effect on prediction accuracy that results from repeated rounds of recombination and the concomitant divergence of LD structure between the training and breeding populations can be alleviated by introducing inbred lines from the product development part as crossing parents in the population improvement part. Such a modified two-part strategy can maintain long-term genetic gain while simultaneously increasing prediction accuracy over time (A. Ceplitis, Lantmännen Lantbruk, Svalöv, Sweden, unpublished data).

The two-part breeding strategy was specifically developed for inbred line crops. Extending the strategy to outcrossing population crops, such as many forage species, which are characterized by significant inbreeding depression and rapid LD decay, is an area of active research. For these types of crops, preliminary results from simulation research indicate that a two-part GS strategy is superior over conventional phenotypic selection and other alternative GS scenarios in terms of accumulated genetic gain, particularly when prediction models include dominance effects (A. Ceplitis, Lantmännen Lantbruk, Svalöv, Sweden, unpublished data).

OUTLOOK

In this review, we delved into the fundamental building blocks of GP methodology and traced its evolution over more than two decades, illustrating its transformative impact on plant breeding. We elucidated how this methodology plays a pivotal role across various breeding stages, aiding in the selection of superior candidate individuals for further crossing, all while minimizing or even eliminating the need for extensive phenotyping over many consecutive breeding generations. This comprehensive review underscores the transformative impact of GP on the enhancement of crop genetic improvement, particularly in revolutionizing cross-breeding. The utilization of high-throughput genomic technology enables a thorough analysis of the entire crop genome, facilitating the identification of promising breeding germplasm associated with desirable traits for subsequent selection. By leveraging extensive sets of genomic and phenotypic data, GS methods predict breeding values for specific traits, thus circumventing the need for laborious and resource-intensive field trials. This streamlined approach speeds up the breeding process, thereby facilitating the selection of superior germplasm with coveted attributes such as increased yield, resistance to pathogens and pests, and adaptability to the dynamic environmental changes, which are often exacerbated by ongoing global warming.

A pivotal strength of GP lies in its capacity to unravel the intricate genetic architecture of traits. In contrast to cross-breeding methods heavily reliant on phenotypic observations influenced by both genetic and environmental factors, GP delves directly into the genetic makeup of plants, offering a more precise and reliable evaluation of their potential performance. This not only simplifies the identification of favorable alleles but also enables plant breeders to consider gene interactions and environmental influences in the target trait(s), thus resulting in the development of more robust and resilient crop germplasm. From this improved

germplasm pool, the selection and further release of desired cultivars become more targeted and effective. As GP of breeding values progresses, its integration with machine learning and artificial intelligence emerges as a promising frontier in crop genetic improvement. The synergy of extensive genomic data and advanced computational models allows for the discernment of subtle genetic patterns and interactions previously overlooked. This holistic approach opens avenues for enhancing crop productivity, sustainability, and resilience in the face of challenges such as climate change and global food and nutrition security. Ultimately, GP of breeding values stands as a cutting-edge approach empowering plant breeders to make informed decisions, thus promising a new wave of innovation in agriculture.

This review explored the impact of various factors on the accuracy of GP with empirical research on wheat, maize, and potato as examples of different reproduction systems. We emphasized that GP, as a predictive tool, relies on the assurance of consistently high or, at the very least, commendable prediction accuracy. Nevertheless, it is worth noting that achieving such precision is not always feasible, owing to the myriad factors that influence its efficacy. We elucidated these factors and offered insights into how they can be optimized to enhance the practical application of GP methodology. Moreover, we expound upon how GS can harness the integration of omics and environmental data to further enhance its accuracy, broadening its scope and applicability. In conclusion, our review underscores that GS can significantly elevate genetic gains per unit of time within crop-breeding programs, but to increase its efficiency it is of paramount importance to integrate all factors that affect GP methodology to fully harness the potential of this groundbreaking predictive data-driven approach.

SUPPLEMENTAL INFORMATION

Supplemental information is available at *Molecular Plant Online*.

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AUTHOR CONTRIBUTIONS

A.A. and A. Chawade conceived the study. A.A., J.Å., O.A.M.-L., J.I.y.S., J.F.-G., W.T., J.C., A. Ceplitis, and R.O. contributed to writing the original manuscript draft; A.A., J.I.y.S., and J.F.-G. created figures; A. Chawade, A.S.C., R.R.V., A. Ceplitis, and R.O. reviewed and edited the manuscript. All authors approved the final manuscript.

DECLARATION OF INTERESTS

A. Ceplitis and J.Å. are employed by Lantmännen Lantbruk.

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